Lab 2: OKE Custom Create with Managed Nodes

Use an existing VCN, custom subnets, pod networking and managed node pool settings

Setting	Value
Region	ap-hyderabad-1 (Hyderabad)
Compartment	Root tenancy compartment
Workflow	Oracle Kubernetes Engine (OKE)
Node type	Managed nodes

Before you begin: Make sure you have permission to create OKE clusters, Compute instances, VCN resources, subnets, load balancers and related networking resources.

Use this workflow when you want to select an existing VCN, subnets, networking type, node placement, security settings or capacity options.

A. Prepare the network first

Before starting Custom Create, prepare or identify the following resources:

- Existing VCN
- Kubernetes API endpoint subnet
- Worker node subnet
- Load balancer subnet
- Pod subnet, if using VCN-native pod networking
- Route tables
- NAT gateway for private worker nodes
- Internet gateway, if public access is required
- Security lists or Network Security Groups (NSGs)
- Optional bastion subnet for private cluster access

Recommended subnet design (basic production-style layout)

Purpose	Recommended configuration
API endpoint	Regional subnet; public or private
Managed worker nodes	Private regional subnet
Pods	Private regional subnet when using VCN-native networking
Load balancers	Regional public subnet for internet-facing applications
Bastion host	Separate public subnet, if required

Note: Make sure the subnet CIDR ranges do not overlap.

B. Start Custom Create

1. Open **Developer Services > Containers & Artifacts > Kubernetes Clusters (OKE)**.
2. Select the target compartment.
3. Click **Create cluster**.
4. Select **Custom create**.
5. Click **Proceed**.

C. Configure basic cluster settings

On the cluster configuration page, enter:

- Cluster name
- Compartment
- Kubernetes version
- Kubernetes API endpoint type: Public or Private
- Optional advanced settings: image verification, Kubernetes secrets encryption, Vault encryption key, cluster add-ons, OIDC discovery, tags

Click **Next**.

D. Configure network resources

On the Network setup page, select the networking model.

Option 1: VCN-native pod networking

Choose this when pods should receive IP addresses from OCI VCN pod subnets.

- Pods are directly routable within the VCN.
- Pod traffic can be integrated with VCN routing.
- Useful for communication with peered VCNs or on-premises networks.

Configure:

1. Select the existing VCN.
2. Select the Kubernetes API endpoint subnet.
3. Select the load balancer subnet.
4. Select the pod subnet, which should normally be private and regional.
5. Select or configure Network Security Groups.
6. Verify that the pod subnet CIDR does not conflict with the worker node subnet.

Option 2: Flannel overlay networking

Choose this when you want the traditional Kubernetes overlay network.

1. Select the existing VCN.
2. Select the API endpoint subnet.
3. Select the worker node subnet.
4. Select the load balancer subnet.
5. Specify the pod CIDR if the Console requests it.
6. Ensure that the pod CIDR does not overlap with the VCN or subnet CIDRs.

Click **Next**.

E. Configure the managed node pool

On the Node pools page:

1. Click **Add node pool** if a node pool is not already displayed.
2. Enter a node pool name, for example **managed-pool-1**.
3. Set **Node type** to **Managed**.
4. Select the Kubernetes version.
5. Select the Compute shape.
6. Select an OKE-compatible image.
7. Enter the desired node count.

Configure node placement

- Availability domain
- Fault domain, if required
- Worker node subnet
- Public or private worker node placement

> **Note:** For production, distribute nodes across fault domains when the region and shape support it.

Configure VCN-native pod networking

If you selected VCN-native pod networking:

1. Select the pod subnet.
2. Confirm that it is private and regional.
3. Select NSGs for worker nodes and for pods.
4. Verify that worker and pod subnet rules allow the required cluster traffic.

Configure optional node-pool settings

Depending on your requirements, configure:

- Capacity type
- Capacity reservation
- Compute host group
- Compute cluster
- Boot volume size
- Boot volume encryption
- SSH public key
- Node labels
- Node taints
- Cloud-init or custom initialization script
- VNIC configuration
- NSGs

Click **Next**.

F. Review and create

1. Review the cluster configuration and confirm: correct region, compartment, VCN, API endpoint subnet, worker node subnet, load balancer subnet, pod networking type, Managed node type, and node count and shape.
2. Click **Create cluster**.
3. Monitor the cluster status.

Wait until the cluster is **Active**, the node pool is **Active**, and the nodes are **Ready**.

G. After the cluster is created

1. Open the cluster.
2. Go to **Node pools**.
3. Confirm the managed node pool is active.
4. Open the node pool and verify that the nodes are running.
5. Go to **Access cluster**.
6. Select the preferred access method (see Lab 1, Part C for Cloud Shell access).
7. Follow the displayed instructions to configure kubectl.
8. Verify the cluster:

```
kubectl get nodes
```

The output should show the managed worker nodes in the Ready state. You can also verify the system pods:

```
kubectl get pods -A
```

H. Recommended Custom Create demo configuration

- Existing VCN
- Private worker node subnet
- Public load balancer subnet
- Private API endpoint or controlled public endpoint
- VCN-native pod networking
- One managed node pool
- Two worker nodes if quota permits
- NSGs for workers and pods
- NAT gateway for private-node outbound access

This configuration is more appropriate for demonstrating production-style networking.

Important checks before clicking Create

- The selected Compute shape is available in ap-hyderabad-1.
- Your trial tenancy has sufficient Compute and networking quota.
- Subnet CIDR ranges do not overlap.
- Private worker nodes have outbound access through a NAT gateway.
- The load balancer subnet is suitable for the type of application traffic you expect.
- You have saved the SSH public key if node access is required.
- Required IAM permissions are available.

Sources

https://docs.oracle.com/en-us/iaas/Content/ContEng/Tasks/contengcreatingclusterusingoke.htm
https://docs.oracle.com/en-us/iaas/Content/ContEng/Concepts/contengnetworkconfig.htm